CSCI2750 Homework 2

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1 Problem 1

1.1 Part a

In $\mathbb{R}^2$, take the instance $p = (1, 0)$ and the dataset

$$x^{(1)} = (1, 1), \qquad x^{(2)} = (3, 0).$$

Consider $x^\star$.

- $d(x^{(1)}, p) = \|x^{(1)} - p\|_2 = \|(0, 1)\|_2 = 1.$
- $d(x^{(2)}, p) = \|x^{(2)} - p\|_2 = \|(2, 0)\|_2 = 2.$

So $d(x^{(1)}, p) < d(x^{(2)}, p)$, hence $x^\star = x^{(1)}$.

Consider $x^\dagger$.

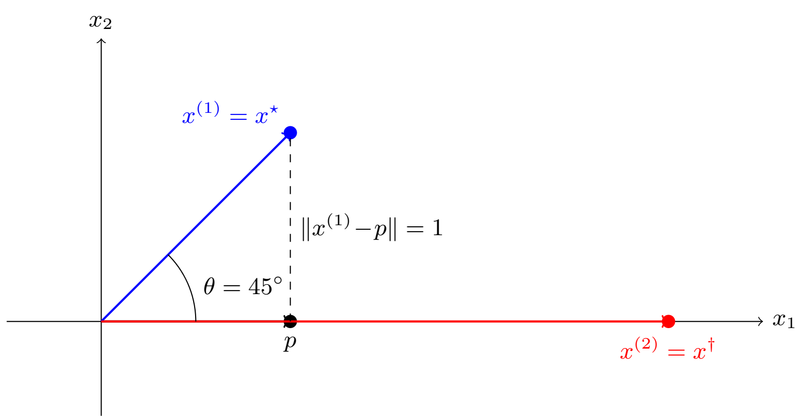
The angle $g(a, b) = \arccos\left(\frac{a^\top b}{\|a\|_2 \|b\|_2}\right)$ is minimized when the vectors are most aligned (smallest angle), trivial to see.

- $x^{(2)}$ and p are both on the positive x-axis, so $g(x^{(2)}, p) = \arccos(1) = 0$.
- $x^{(1)}$ makes angle $\theta = 45°$ with p, trivial to see.

So $g(x^{(2)}, p) < g(x^{(1)}, p)$, hence $x^\dagger = x^{(2)}$.

Thus $x^\star = x^{(1)} \neq x^{(2)} = x^\dagger$.

Geometric illustration:



1.2 Part (b)

For $\|p\|_2 = \|x^{(i)}\|_2 = 1$ for all $i = 1, \dots, m$, all possible vectors of $x^{(i)}$ lies on the unit circle with center at the origin.

$$d(x,p)^2 = \|x - p\|_2^2 = \|x\|_2^2 + \|p\|_2^2 - 2\,x^\top p = 2(1 - x^\top p).$$

So we need to maximize $x^\top p$ in order to minimize $d(x, p)$.

For unit vectors, $g(x, p) = \arccos\left(\frac{x^\top p}{\|x\|_2\|p\|_2}\right) = \arccos(x^\top p)$.
Since arccos is a decreasing function, minimizing $g(x, p)$ is equivalent to maximizing $x^\top p$ and minimize $d(x, p)$.

Therefore, with the smallest x^* possible, we will also have the smallest $x^\dagger$ possible.

2 Problem 2

2.1 Part a

Manhattan distance: $d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|$.

- Non-negativity: Manhattan distance means sum of absolute differences(each element is non-negative), so it is always non-negative.
- identity of indiscernibles: For $d_1(x, y) = 0$, to make this happen, all corresponding elements are the same. i.e. $x = y$.
- Symmetry: $d_1(x, y) = d_1(y, x)$ because for absolute value, $|a - b| = |b - a|$.
- Triangle inequality: For each i, the triangle inequality for real numbers gives $|x_i - z_i| \le |x_i - y_i| + |y_i - z_i|$. Summing over i:

  $$d_1(x, z) = \sum_{i=1}^{n} |x_i - z_i| \le \sum_{i=1}^{n} |x_i - y_i| + \sum_{i=1}^{n} |y_i - z_i| = d_1(x, y) + d_1(y, z).$$

  Therefore d_1 is a metric.

2.2 Part b

For each k, convexity of f_k gives

$$f_k(\alpha x_1 + (1 - \alpha)x_2) \le \alpha f_k(x_1) + (1 - \alpha) f_k(x_2).$$

Multiply by c_k, Assume $c_k \ge 0$ for all k to preserve the inequality:

$$c_k f_k(\alpha x_1 + (1 - \alpha)x_2) \le \alpha c_k f_k(x_1) + (1 - \alpha) c_k f_k(x_2).$$

Sum over for $k = 1$ to K:

$$\sum_{k=1}^{K} c_k f_k(\alpha x_1 + (1 - \alpha)x_2) \le \alpha \sum_{k=1}^{K} c_k f_k(x_1) + (1 - \alpha) \sum_{k=1}^{K} c_k f_k(x_2).$$

If we define $f(x) = \sum_{k=1}^{K} c_k f_k(x)$, then by definition of f,

$$f(\alpha x_1 + (1 - \alpha)x_2) \le \alpha f(x_1) + (1 - \alpha) f(x_2).$$

Therefore f is convex.

2.3 Part c

$$x^\top Gx - 2r^\top Ax + \|r\|_2^2 = x^\top A^\top Ax - 2r^\top Ax + \|r\|_2^2 = \|Ax - r\|_2^2.$$

Therefore, we have:

$$f(x) = e^{x^\top Gx - 2r^\top Ax + \|r\|_2^2} = e^{\|Ax-r\|_2^2}.$$

We can write f as a composition of three convex functions s.t. $f = f_1(f_2(f_3(x)))$, listing from inside to outside:

- $f_3(x) = Ax - r$, with $f_3 : \mathbb{R}^n \to \mathbb{R}^n$ (affine function);
- $f_2(x) = \|x\|_2^2$, with $f_2 : \mathbb{R}^n \to \mathbb{R}$ (squared norm, convex);
- $f_1(x) = e^x$, with $f_1 : \mathbb{R} \to \mathbb{R}$ (strictly convex because exponential, non-decreasing).

By convex-preserving rules...

$f_2(f_3(x)) = \|Ax - r\|_2^2$ is convex because it is the composition of a convex function with an affine function.

$f_1(f_2(f_3(x))) = e^{\|Ax-r\|_2^2}$ is convex because it is the composition of a non-decreasing convex function with another convex function.

Therefore f is convex.

3 Problem 3

3.1 Part a

Both terms in f are non-negative and are thus minimized when $x_1 = 0$ and $x_2 = 0$, giving $f(0,0) = 0$.

Also, The point $(0,0)$ satisfies the constraint $x_1^2 + x_2^2 \leq 1$, so the solution is feasible as well.

$$x^\star = (0,0), \qquad f(x^\star) = 0.$$

3.2 Part b

We write (P) in the standard convex form

$$\min_{x\in\mathbb{R}^2} f(x) \quad \text{s.t.} \quad g(x) \leq 0,$$

where

$$f(x) = \frac{1}{4}x_1^4 + \frac{3}{2}x_2^2, \qquad g(x) = x_1^2 + x_2^2 - 1.$$

$f(x)$ is convex because:

- The function x^2 is convex, so x_1^2 and x_2^2 are convex functions of x_1 and x_2 respectively.
- The function x^4 can be written as $x^4 = (x^2)^2$. x^2 is convex and non-decreasing for $x \geq 0$. For the inner x^2, it must be non-negative, so we can see the outer function is non-decreasing. It is a composition of a convex non-decreasing convex function with another convex function, hence x^4 and thus x_1^4 are convex.
- Both x_1^4 and x_2^2 is scaled by a positive constant, and sumed up to form $f(x)$. In this case, the convexity is preserved.

3.3 Part c

$$\mathcal{L}(x_1, x_2, \alpha) = \frac{1}{4}x_1^4 + \frac{3}{2}x_2^2 + \alpha\,(x_1^2 + x_2^2 - 1).$$

Consider the KKT conditions:

- **Stationarity:** take derivatives of $\mathcal{L}$ with respect to x_1 and x_2 and set them to zero:

$$\frac{\partial \mathcal{L}}{\partial x_1} = x_1^3 + 2\alpha x_1 = 0 \quad \Rightarrow \quad x_1(x_1^2 + 2\alpha) = 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 3x_2 + 2\alpha x_2 = 0 \quad \Rightarrow \quad x_2(3 + 2\alpha) = 0.$$

- **Primal feasibility:**

$$g(x) = x_1^2 + x_2^2 - 1 \leq 0.$$

- **Dual feasibility:**

$$\alpha \geq 0.$$

- **Complementary slackness:**

$$\alpha\,(x_1^2 + x_2^2 - 1) = 0.$$

For $\alpha \geq 0$, we have no solution for $x_1^2 + 2\alpha = 0$ and $3 + 2\alpha = 0$, so we must have

$$x_1 = 0, \quad x_2 = 0.$$

Then Complementary slackness gives $\alpha(x_1^2 + x_2^2 - 1) = 0$. Since $x_1 = 0$ and $x_2 = 0$, we have $x_1^2 + x_2^2 - 1 = -1 < 0$, forcing $\alpha = 0$.

Thus the KKT point is $(x_1^\star, x_2^\star, \alpha^\star) = (0, 0, 0)$, consistent with the solution found in Part (a).

3.4 Part d

In MATLAB, I used the following code to solve the problem:

```
cvx_begin
    variables x1 x2
    minimize( 0.25 * pow_pos(x1, 4) + 1.5 * square(x2) )
    subject to
        square(x1) + square(x2) <= 1;
cvx_end
```

The result is "Optimal value (cvx_optval): +3.20163e-12", essentially the same as the solution found in Part (a) with a extremely small numerical error.

4 Problem 4

4.1 list of wikipages

Five modern scientists:

- Stephen Hawking: `https://en.wikipedia.org/wiki/Stephen_Hawking`
- James Watson: `https://en.wikipedia.org/wiki/James_Watson`
- Tu Youyou: `https://en.wikipedia.org/wiki/Tu_Youyou`
- Albert Einstein: `https://en.wikipedia.org/wiki/Albert_Einstein`

- Timothy Berners-Lee: `https://en.wikipedia.org/wiki/Timothy_Berners-Lee`

Five athletes:

- Kobe Bryant: `https://en.wikipedia.org/wiki/Kobe_Bryant`
- Cristiano Ronaldo: `https://en.wikipedia.org/wiki/Cristiano_Ronaldo`
- Lionel Messi: `https://en.wikipedia.org/wiki/Lionel_Messi`
- Michael Phelps: `https://en.wikipedia.org/wiki/Michael_Phelps`
- Usain Bolt: `https://en.wikipedia.org/wiki/Usain_Bolt`

all pages are accessed on 2026-02-20. 12:00nn.

4.2 Dictionary

I chose the following words as the dictionary:

- Technology
- Speed
- Death
- Physics
- Olympic
- Rival
- Champion

4.3 Word occurrence counts

Counts of the 7 dictionary words on each Wikipedia page:

Page	Technology	Speed	Death	Physics	Olympic	Rival	Champion
Stephen Hawking	18	2	46	107	0	0	1
James Watson	8	0	7	6	0	0	0
Tu Youyou	18	0	1	4	2	0	0
Albert Einstein	12	7	19	189	0	0	0
Timothy Berners-Lee	34	0	0	3	1	0	0
Kobe Bryant	1	1	62	0	38	1	6
Cristiano Ronaldo	0	1	4	0	7	1	3
Lionel Messi	0	3	1	0	15	12	19
Michael Phelps	1	1	2	0	326	5	11
Usain Bolt	9	24	0	3	177	8	26
Total	101	39	142	312	566	27	66

4.4 Heatmap

By using MATLAB, I generated the following heatmap:

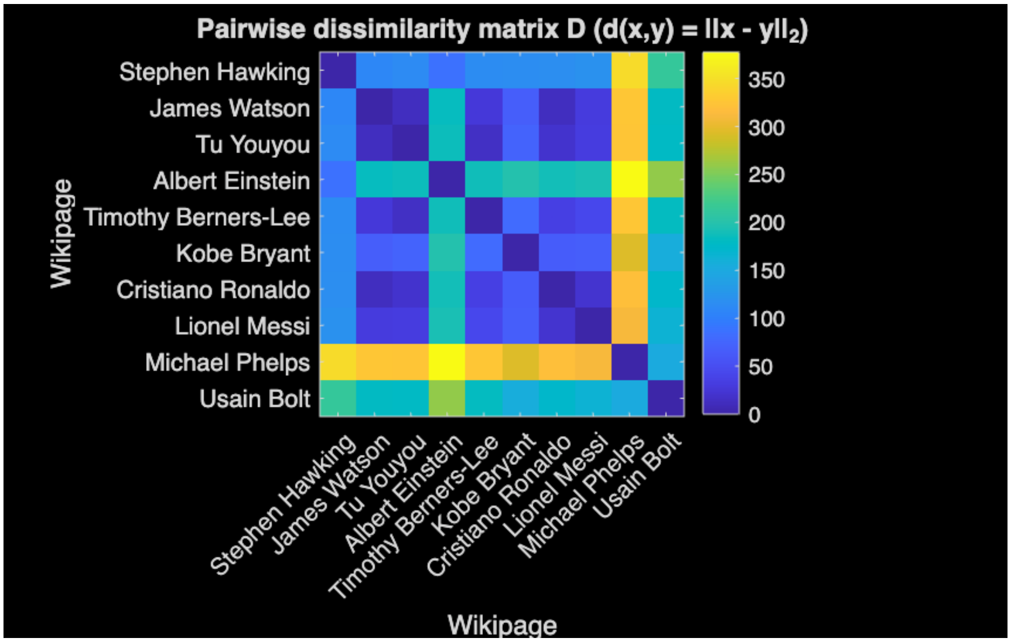


Figure 1: Heatmap of the word occurrence counts

4.5 Brief analyis

We can make use of the euclidean distance of count vector to measure the similarity between two pages. The smaller the distance, the more similar the two pages are.
There for we can group the pages into different clusters based on the distance.
A heatmap is a good way to visualize the similarity between two pages.

4.6 Appendix

Below is the MATLAB code used to generate the heatmap:

```
% Word-count vectors
X = [18  2 46 107   0   0  1;   % Stephen Hawking
8  0  7   6   0   0  0;   % James Watson
18  0  1   4   2   0  0;   % Tu Youyou
12  7 19 189   0   0  0;   % Albert Einstein
34  0  0   3   1   0  0;   % Timothy Berners-Lee
1  1 62   0  38   1  6;   % Kobe Bryant
0  1  4   0   7   1  3;   % Cristiano Ronaldo
0  3  1   0  15  12 19;   % Lionel Messi
1  1  2   0 326   5 11;   % Michael Phelps
9 24  0   3 177   8 26];  % Usain Bolt

names = {'Stephen Hawking', 'James Watson', 'Tu Youyou', 'Albert Einstein', ...
         'Timothy Berners-Lee', 'Kobe Bryant', 'Cristiano Ronaldo', ...
         'Lionel Messi', 'Michael Phelps', 'Usain Bolt'};
n = size(X, 1);
D = zeros(n, n);
for i = 1:n
    for j = 1:n
        diff = X(i,:) - X(j,:);
        D(i,j) = sqrt(sum(diff.^2));
    end
end

% Heatmap
figure;
imagesc(D);
colorbar;
title('Pairwise dissimilarity matrix D (d(x,y) = ||x - y||_2)');
xlabel('Wikipage');
ylabel('Wikipage');
set(gca, 'XTick', 1:n, 'XTickLabel', names, 'XTickLabelRotation', 45);
set(gca, 'YTick', 1:n, 'YTickLabel', names);
axis square;

saveas(gcf, 'heatmap.png');
```